Banknote Authenticity Detection using Computer Vision

1. Introduction

Counterfeit banknotes represent a serious problem for modern economies. Every year, financial institutions and commercial activities lose large amounts of money due to fake currency circulation. Human inspection is often insufficient because counterfeit techniques are becoming increasingly sophisticated, making fake banknotes visually similar to genuine ones. Computer Vision and Machine Learning provide an effective way to automate the authentication process by analyzing visual patterns that may not be immediately visible to the human eye. The goal of this project is to design and implement an automated system capable of classifying banknotes as:

- **Genuine**
- **Fake** using two different Artificial Intelligence approaches:

1. A **Classical Computer Vision + Machine Learning** pipeline
2. A **Deep Learning** pipeline based on Convolutional Neural Networks (CNNs)

The project also aims to compare the strengths and weaknesses of both approaches on a relatively small dataset.

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2. Objectives of the Project

The main objectives are:

- Acquire and preprocess banknote images
- Extract meaningful visual information
- Train classification models
- Evaluate model performance using standard metrics
- Compare traditional Computer Vision techniques with Deep Learning methods
- Understand the limitations of small datasets in AI systems The project was designed not only as a practical implementation but also as a study of how different Computer Vision methodologies behave in real-world classification problems.

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3. Dataset Description

The dataset contains images of both genuine and fake banknotes. The images are divided into two main categories:

- `genuine`
- `fake`
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The dataset is further separated into:

- Training set
- Test set

The banknotes include different euro denominations such as:

- 10€
- 20€
- 50€

3.1 Dataset Challenges

The dataset presents several important challenges:

Small Size

The number of available images is relatively limited. This strongly affects Deep Learning models because CNNs typically require thousands of examples.

Lighting Variations

Some images contain different illumination conditions:

- Shadows
- Reflections
- Uneven brightness This complicates feature extraction.

Perspective Distortions

Banknotes are not always perfectly aligned with the camera.

Artificial Counterfeits

Some fake banknotes are artificially generated or simulated, meaning they may not perfectly represent real counterfeit money.

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4. Classical Computer Vision Approach

The first approach uses traditional Computer Vision techniques combined with Machine Learning. This pipeline follows several sequential steps.

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5. Image Preprocessing

Preprocessing is one of the most important phases because raw images often contain noise and irrelevant information. The goal is to normalize the images before feature extraction.

5.1 Grayscale Conversion

The RGB image is converted into grayscale. Why? Because many structural features do not require color information. Advantages:

- Reduces computational complexity
- Simplifies feature extraction
- Removes irrelevant chromatic variations A grayscale image contains only intensity values.

5.2 Gaussian Blur

Gaussian Blur is applied to reduce noise. Noise may originate from:

- Camera sensors
- Compression artifacts
- Illumination irregularities The Gaussian filter smooths the image by averaging nearby pixel values according to a Gaussian distribution. This improves edge detection stability.

5.3 Edge Detection

Edges represent abrupt intensity changes. In Computer Vision, edges are extremely important because they describe object boundaries and structural information. The project uses edge detection to identify:

- Borders
- Printed patterns
- Shapes
- Security structures

Common edge detectors include:

- Sobel
- Canny
- Laplacian Edge detection transforms the image into a representation emphasizing contours and geometric structures.

5.4 Contour Detection

Contours are continuous curves representing object boundaries. Contour detection allows the system to isolate the banknote from the background. This is important because:

- Background pixels introduce noise
- Classification should focus only on the banknote The largest contour usually corresponds to the banknote itself.

5.5 Perspective Transformation

Banknotes may appear rotated or tilted. Perspective transformation corrects the orientation by mapping the detected contour into a rectangular shape. Benefits:

- Standardized image dimensions
- Improved feature consistency
- Better classifier performance Without perspective correction, extracted features could vary significantly between images.

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6. Feature Extraction

Feature extraction transforms images into numerical vectors understandable by Machine Learning algorithms. The project combines multiple descriptors.

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7. Histogram of Oriented Gradients (HOG)

HOG is one of the most important feature descriptors in Computer Vision. It captures shape and structural information. The algorithm works by:

1. Computing image gradients
2. Dividing the image into cells
3. Building orientation histograms
4. Normalizing blocks

HOG is particularly effective for:

- Printed patterns
- Symbols
- Geometric structures
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In banknotes, HOG captures:

- Text structures
- Borders
- Fine geometric details

The resulting descriptor is a high-dimensional feature vector.

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8. Local Binary Patterns (LBP)

LBP is a texture descriptor. It analyzes local neighborhoods of pixels. For each pixel:

- Neighboring pixels are compared to the center
- Binary values are generated
- A binary pattern is constructed

LBP is very useful for detecting texture inconsistencies. Why is this important? Counterfeit banknotes often fail to reproduce microscopic textures accurately. LBP helps detect:

- Printing irregularities
- Surface differences
- Texture anomalies

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9. Color Histograms

Although grayscale is heavily used, color information may still contain useful signals. Color histograms represent the distribution of pixel intensities. They help analyze:

- Ink distributions
- Color consistency
- Tonal variations

Fake banknotes may exhibit abnormal color distributions due to lower printing quality.

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10. Statistical Features

Additional statistical descriptors are extracted, such as:

- Mean intensity
- Standard deviation
- Variance
- Entropy

These features summarize global image properties. For example:

- Genuine banknotes may show more consistent variance patterns
- Fake banknotes may contain irregular intensity distributions

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11. Feature Vector Construction

All extracted descriptors are concatenated into a single numerical vector. This vector becomes the input of the classifier. The process transforms a complex image into structured numerical information suitable for Machine Learning.

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12. Support Vector Machine (SVM)

The classical approach uses a Support Vector Machine classifier. SVM is a supervised Machine Learning algorithm designed for classification tasks.

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13. How SVM Works

SVM attempts to find the optimal hyperplane separating two classes. The optimal hyperplane maximizes the margin between classes. Key concepts:

- Support vectors
- Margin maximization
- Decision boundary The classifier learns from labeled examples.

In this project:

- Genuine banknotes → class 1
- Fake banknotes → class 0

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14. Why SVM Was Chosen

SVM performs extremely well on:

- Small datasets
- High-dimensional feature spaces
- Structured features like HOG

Advantages include:

- Good generalization
- Robustness
- Efficient performance with limited data

This makes SVM ideal for the project scenario.

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15. Deep Learning Approach

The second approach uses Convolutional Neural Networks. Unlike classical methods, CNNs automatically learn features from images. No manual feature engineering is required.

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16. Convolutional Neural Networks (CNNs)

CNNs are specialized neural networks for image processing. They are inspired by the human visual cortex. A CNN learns hierarchical features:

- Early layers → edges
- Intermediate layers → textures
- Deep layers → semantic patterns

CNNs are extremely powerful for image classification tasks.

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17. MobileNetV2 Architecture

The project uses **MobileNetV2**. MobileNetV2 is:

- Lightweight
- Efficient
- Optimized for limited computational resources

It was originally trained on ImageNet.

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18. Transfer Learning

The project uses Transfer Learning. Instead of training from scratch:

- Pretrained weights are reused
- Knowledge learned from ImageNet is transferred Benefits:
- Faster training
- Better generalization
- Reduced data requirements

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19. Frozen Layers

Most MobileNetV2 layers are frozen. This means their weights are not updated during training. Only the final classification layer is replaced and trained for:

- Genuine
- Fake This reduces overfitting on small datasets.

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20. Data Augmentation

CNN performance improves with data augmentation. Possible augmentations include:

- Rotation
- Flipping
- Brightness changes
- Zoom
- Translation

Data augmentation artificially increases dataset diversity. This helps improve generalization.

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21. Training Process

The CNN training process involves:

1. Forward propagation
2. Loss computation
3. Backpropagation
4. Weight optimization The optimizer minimizes classification error over multiple epochs.

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22. Evaluation Metrics

The models are evaluated using standard metrics.

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23. Accuracy

Accuracy measures overall correctness. Formula: Accuracy = \frac{TP + TN}{TP + TN + FP + FN} Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

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24. Precision

Precision measures prediction reliability. Precision = \frac{TP}{TP + FP} High precision means few false positives.

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25. Recall

Recall measures the ability to detect positive samples. Recall = \frac{TP}{TP + FN} High recall means few false negatives.

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26. F1-Score

F1-score balances precision and recall. F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}

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27. Confusion Matrix

The confusion matrix summarizes predictions. It contains:

- True Positives
- True Negatives
- False Positives
- False Negatives

This helps understand model behavior in detail. For counterfeit detection:

- False positives may reject genuine money
- False negatives may accept counterfeit money

Both errors are important.

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28. Experimental Results

The project compares:

- SVM
- CNN (MobileNetV2)

using:

```
python src/compare_models.py
```

The script generates:

- `comparison_svm_vs_cnn.txt`
- `comparison_table.png`

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29. Analysis of Results

The SVM model achieved better performance than the CNN. This may initially seem surprising because Deep Learning is generally considered superior. However, this result is completely coherent with Machine Learning theory.

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30. Why SVM Performed Better

Several factors explain the results.

30.1 Small Dataset

CNNs require large datasets. With insufficient data:

- CNNs overfit easily
- Generalization becomes weak SVM handles small datasets more effectively.

30.2 Handcrafted Features

HOG and LBP provide strong engineered features. They already encode relevant visual information. The SVM classifier receives high-quality structured inputs.

30.3 Computational Simplicity

SVM training is lighter and more stable. CNN training requires:

- More epochs
- More GPU power
- Larger datasets

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31. Failure Analysis

The system still has limitations.

31.1 Lighting Sensitivity

Different lighting conditions affect feature extraction. Reflections and shadows may confuse the classifier.

31.2 Real Counterfeit Diversity

Real counterfeit techniques vary significantly. The dataset cannot represent all possible fake banknotes.

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32. Ethical Considerations

AI systems must be evaluated critically. This project should not be used in real banking systems without extensive validation. Possible risks include:

- Wrong accusations
- Financial losses
- Bias from unbalanced datasets Human supervision remains essential.

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33. Possible Future Improvements

Several improvements could significantly enhance the project.

33.1 Larger Dataset

More data would dramatically improve CNN performance.

33.2 Real Counterfeit Samples

Using real fake banknotes would increase realism.

33.3 Advanced Architectures

More advanced CNNs could be tested:

- EfficientNet
- ResNet
- Vision Transformers

33.4 Real-Time Detection

The system could be integrated with:

- Mobile cameras
- ATM systems
- Embedded devices

33.5 Explainable AI

Visualization techniques like Grad-CAM could explain which regions influenced predictions.

This improves interpretability.

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34. Conclusion

This project demonstrates how Computer Vision can solve real-world classification problems. Two different paradigms were explored:

1. Classical Computer Vision + SVM
2. Deep Learning with CNNs

The experiments show that:

- Traditional methods remain extremely effective on small datasets
- Feature engineering still plays an important role
- CNNs require significantly larger datasets
- Transfer Learning helps but cannot fully compensate for limited data

The project also highlights the importance of:

- Preprocessing
- Feature extraction
- Dataset quality
- Proper evaluation metrics

Overall, the system represents a complete Computer Vision pipeline combining:

- Image processing
- Machine Learning
- Deep Learning
- Experimental evaluation
- Comparative analysis

and provides a strong practical example of modern AI-based image classification systems.